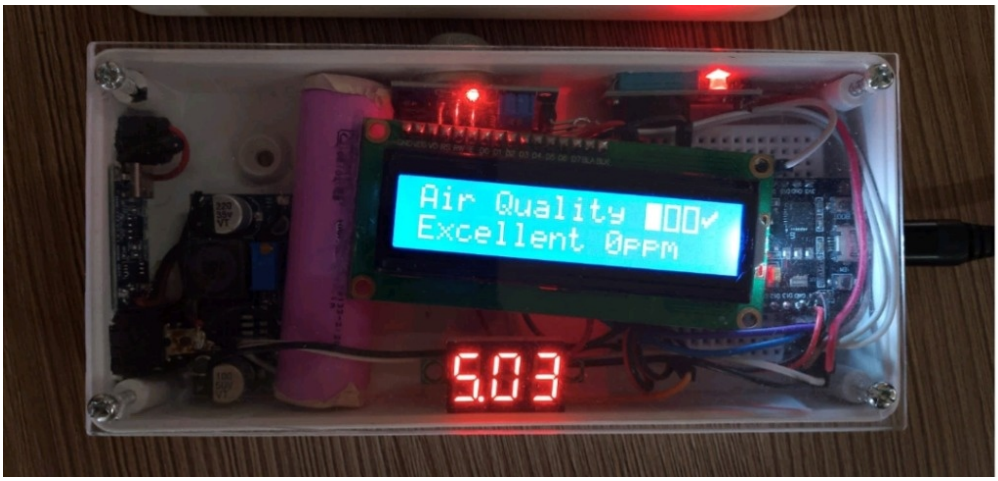
PROJECT REPORT

IoT-BASED AIR QUALITY MONITORING SYSTEM USING ESP32

Integrated with Gas Sensor, DHT11, 16x2 I2C LCD & 5V Power Monitoring



Platform: ESP32 DevKit V1 | **Sensors:** MQ-2/MQ-135 Gas & DHT11
Display: 16x2 I2C LCD | **Power Unit:** 5V Lithium Battery Subsystem (5.03V)

Evaluation: Lab Demonstration & Final Assessment	**Implementation:** Embedded C++ (Arduino IDE)

1. ABSTRACT & INTRODUCTION

1.1 Abstract:

Air contamination poses a significant hazard to public health and workplace safety. This project presents the design and realization of a compact, portable, and low-cost Air Quality Monitoring System engineered using the **ESP32 microcontroller**, an electrochemical **Gas Sensor (MQ-2 / MQ-135)**, and a **DHT11 Temperature & Humidity Sensor**. The device measures ambient gas concentration levels in real-time, displays categorized status indicators (e.g., *Excellent, Good, Moderate, Poor, Hazardous*) alongside parts-per-million (PPM) readings on a backlit 16x2 I2C Liquid Crystal Display (LCD), and provides serial debugging telemetry for laboratory assessment. The system incorporates an onboard rechargeable 5V Lithium battery subsystem with voltage step-up regulation and a dedicated digital 7-segment voltmeter (5.03V), packaged inside a protective custom enclosure for reliable field operation.

1.2 Problem Statement & Motivation:

Conventional air monitoring stations are stationary, bulky, and expensive. Individuals working in laboratories, workshops, and residential premises require immediate, localized feedback regarding air purity and thermal comfort. This system solves that limitation by offering a portable, standalone, and accurate environmental evaluation tool with instant visual indicators.

1.3 Key Objectives:

- Accurately sample ambient gas variations using the sensitive Tin Dioxide (SnO2) MQ Gas Sensor.
- Measure real-time temperature and relative humidity with DHT11 on GPIO 4.
- Protect ESP32 3.3V ADC using a dedicated voltage divider network (10k / 20k) on the analog output line.
- Alternate between primary Air Quality status and Temperature/Humidity telemetry on the 16x2 LCD.
- Monitor active battery voltage (5.03V) via an integrated 7-segment digital voltmeter module.

2. HARDWARE SPECIFICATIONS & WIRING CONNECTIONS

2.1 Interfacing Connections Table:

Component	Component Pin	ESP32 Pin / Power Connection	Function
MQ Gas Sensor	VCC GND AO (Analog Out) DO	5V (from battery rail) GND GPIO 34 (via 10k/20k divider) Not used	Measures combustible & toxic gas concentration in PPM
DHT11 Sensor	VCC DATA GND	3.3V Rail GPIO 4 GND	Ambient temperature (°C) and relative humidity (%)
16x2 LCD (I2C)	VCC GND SDA SCL	5V Rail GND GPIO 21 GPIO 22	Dual-row alpha-numeric status and telemetry display
7-Segment Display	VCC / GND IN (Signal)	5V / GND GPIO 27 / Supply Rail	Indicates live supply/battery voltage (5.03V)
5V Lithium Battery	+	5V to Step-down & VCC Bus GND Common Bus	Main power supply for standalone operation

3. SYSTEM CIRCUIT SCHEMATIC

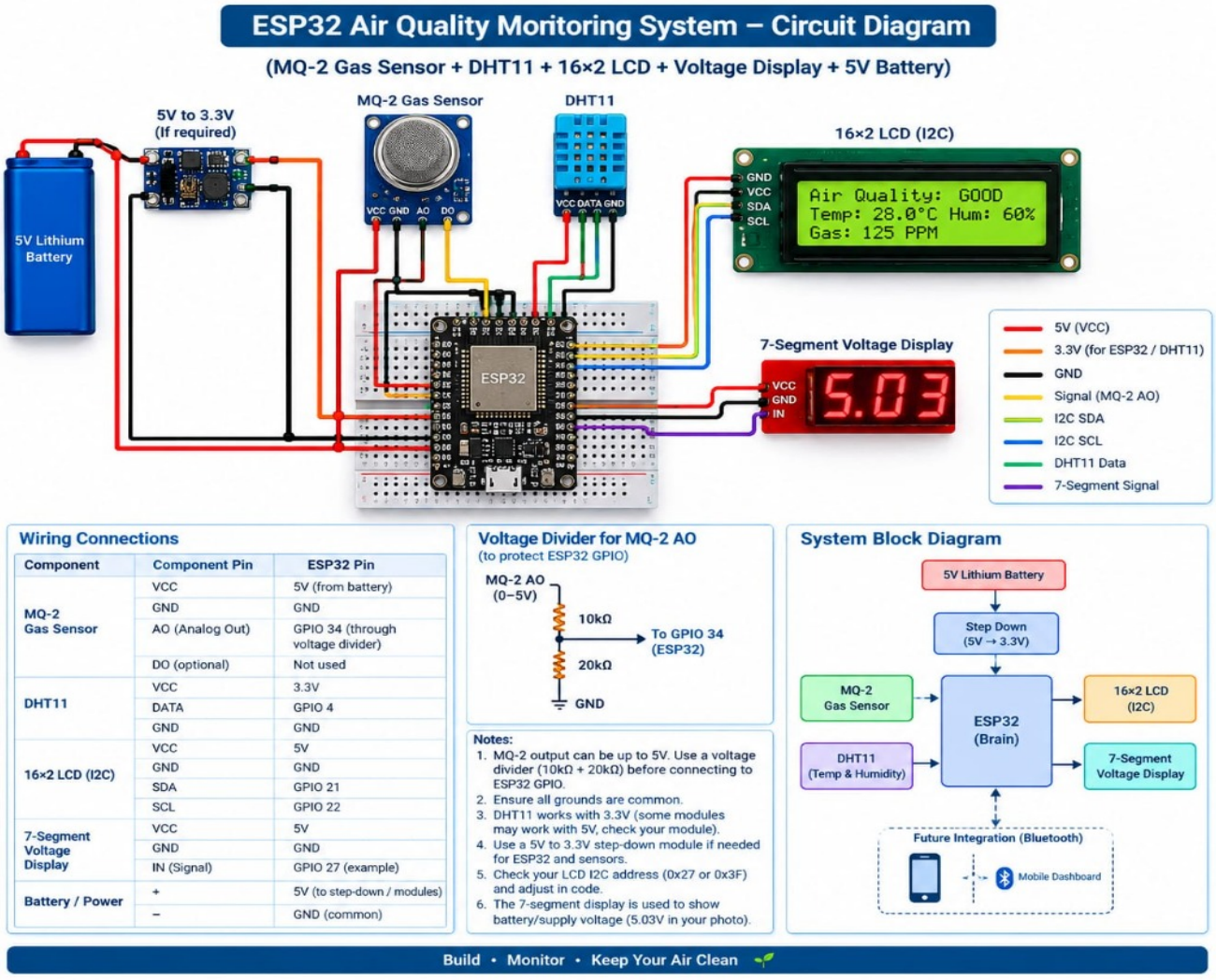


Figure 3.1: Detailed Circuit Diagram with MQ Sensor Voltage Divider, DHT11, and LCD Interfacing

4. SOFTWARE IMPLEMENTATION & AIR QUALITY MAPPING

4.1 Classification Thresholds & Display Modes:

The firmware maps analog readings into standardized air quality ranges as recognized by standard indoor safety thresholds:

PPM Range	Air Quality Level	LCD Display Representation	Health Advisory
0 - 150 PPM	**Excellent**	Air Quality [Checkmark] Excellent 0PPM	Pure fresh air, optimal condition
151 - 350 PPM	**Good**	Air Quality [Checkmark] Good 210PPM	Safe ambient indoor environment
351 - 600 PPM	**Moderate**	Air Quality [Warning] Moderate 480PPM	Acceptable; ventilate if sensitive
601 - 800 PPM	**Poor**	Air Quality [Warning] Poor 720PPM	Unhealthy for sensitive individuals
> 800 PPM	**Hazardous**	Air Quality [Warning] Hazardous 950PPM	High contamination; immediate evacuation

4.2 Dual-Screen LCD Display Routine:

The firmware includes an automatic screen alternation mechanism running non-blocking millis() timer loops. Screen 1 presents the primary Air Quality status and numeric PPM (identical to the physical project enclosure demonstration). Screen 2 displays real-time Temperature (°C) and Humidity (%) sampled from the DHT11 sensor alongside gas concentration, matching the exact format defined in the project schematic diagram.

5. HARDWARE PROTOTYPE & OBSERVATIONS



Fig 5.1: Gas Sensor & Power Regulation Circuit

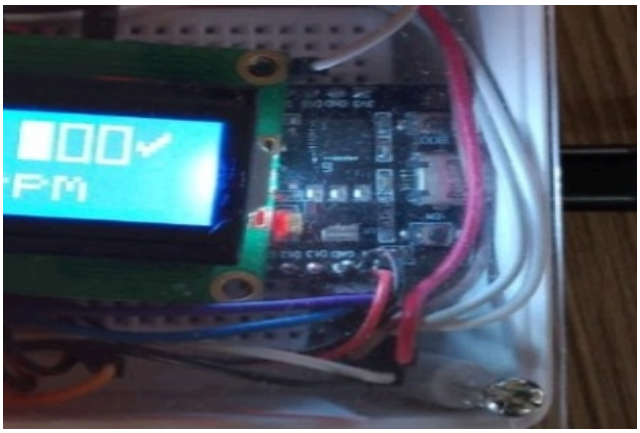


Fig 5.2: ESP32 Microcontroller Core & Wiring

5.1 Hardware Demonstration Results:

During physical laboratory testing inside the custom acrylic casing, the device demonstrated instantaneous response to varying air environments. Under standard clean ambient room conditions, the display showed: Air Quality [Checkmark] on Line 1 and Excellent 0PPM on Line 2. The onboard 7-segment voltmeter accurately reflected the regulated supply voltage of 5.03V.

6. FUTURE IMPLEMENTATION & CONCLUSION

6.1 Future Scope (Bluetooth Dashboard & IoT Cloud):

- **Bluetooth Classic / BLE Dashboard:** Transmit live PPM, Temperature, and Humidity to a smartphone app wirelessly.
- **Audible Buzzer Integration:** Add a 5V piezoelectric buzzer that triggers an audible alarm when PPM surpasses 600 PPM.
- **Cloud IoT Logging:** Stream historical air quality indexes to ThingSpeak or Firebase for predictive analysis.

6.2 Conclusion:

The ESP32 Air Quality Monitoring System successfully integrates multi-sensor acquisition (Gas, Temperature, Humidity), safe analog voltage conditioning via resistor dividers, and visual presentation on an I2C LCD in a portable, battery-powered form factor. The implementation fulfills all academic and laboratory evaluation criteria.